

```
import 'package:cloud_firestore/cloud_firestore.dart';
import 'package:firebase_auth/firebase_auth.dart';
import 'package:firebase_core/firebase_core.dart';
import 'package:flutter/material.dart';
import 'firebase_options.dart';
```

```
Future<void> main() async {
  WidgetsFlutterBinding.ensureInitialized();
  await Firebase.initializeApp(
    options: DefaultFirebaseOptions.currentPlatform,
  );
  runApp(const MyApp());
}
```

```
class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Flutter Firebase Auth App',
      debugShowCheckedModeBanner: false,
      theme: ThemeData(
        useMaterial3: true,
        colorSchemeSeed: Colors.indigo,
      ),
      home: const AuthGate(),
    );
  }
}
```

```
class AuthGate extends StatelessWidget {
  const AuthGate({super.key});

  @override
  Widget build(BuildContext context) {
    return StreamBuilder<User?>(
      stream: FirebaseAuth.instance.authStateChanges(),
      builder: (context, snapshot) {
        if (snapshot.connectionState == ConnectionState.waiting) {
          return const Scaffold(
            body: Center(child: CircularProgressIndicator()),
          );
        }

        if (snapshot.hasData) {
          return const HomeScreen();
        }

        return const LoginScreen();
      }
    );
  }
}
```

```

    },
  );
}
}

```

```

class LoginScreen extends StatefulWidget {
  const LoginScreen({super.key});

  @override
  State<LoginScreen> createState() => _LoginScreenState();
}

```

```

class _LoginScreenState extends State<LoginScreen> {
  final emailController = TextEditingController();
  final passwordController = TextEditingController();

```

```

  bool isLoading = false;
  String errorMessage = '';

```

```

  Future<void> loginUser() async {
    FocusScope.of(context).unfocus();

```

```

    final email = emailController.text.trim();
    final password = passwordController.text.trim();

```

```

    setState(() {
      errorMessage = '';
    });

```

```

    if (email.isEmpty || password.isEmpty) {
      setState(() {
        errorMessage = 'Email and password are required';
      });
      return;
    }

```

```

    setState(() {
      isLoading = true;
    });

```

```

    try {
      await FirebaseAuth.instance.signInWithEmailAndPassword(
        email: email,
        password: password,
      );
    } on FirebaseAuthException catch (e) {
      String msg = 'Login failed';

```

```

      if (e.code == 'invalid-email') {
        msg = 'Invalid email address';

```

```

    } else if (e.code == 'invalid-credential') {
        msg = 'Wrong email or password';
    } else if (e.code == 'user-not-found') {
        msg = 'No user found for this email';
    } else if (e.code == 'wrong-password') {
        msg = 'Wrong password';
    } else if (e.code == 'user-disabled') {
        msg = 'This user account is disabled';
    } else {
        msg = e.message ?? e.code;
    }

    setState(() {
        errorMessage = msg;
    });
} catch (e) {
    setState(() {
        errorMessage = 'Unexpected error: $e';
    });
} finally {
    if (mounted) {
        setState(() {
            isLoading = false;
        });
    }
}
}

class RegisterScreen extends StatefulWidget {
    const RegisterScreen({super.key});

    @override
    State<RegisterScreen> createState() => _RegisterScreenState();
}

class _RegisterScreenState extends State<RegisterScreen> {
    final nameController = TextEditingController();
    final emailController = TextEditingController();
    final passwordController = TextEditingController();

    bool isLoading = false;
    String errorMessage = '';

    Future<void> registerUser() async {
        FocusScope.of(context).unfocus();

        final name = nameController.text.trim();
        final email = emailController.text.trim();
        final password = passwordController.text.trim();

        setState(() {

```

```

    errorMessage = "";
  });

  if (name.isEmpty || email.isEmpty || password.isEmpty) {
    setState(() {
      errorMessage = 'All fields are required';
    });
    return;
  }

  if (!email.contains('@') || !email.contains('.')) {
    setState(() {
      errorMessage = 'Enter a valid email';
    });
    return;
  }

  if (password.length < 6) {
    setState(() {
      errorMessage = 'Password must be at least 6 characters';
    });
    return;
  }

  setState(() {
    isLoading = true;
  });

  final user = credential.user;
  if (user == null) {
    throw Exception('User creation failed');
  }

  await user.updateDisplayName(name);
  // Write profile document
  await FirebaseFirestore.instance.collection('users').doc(user.uid).set({
    'uid': user.uid,
    'name': name,
    'email': email,
    'createdAt': FieldValue.serverTimestamp(),
  }, SetOptions(merge: true));
  // No Navigator needed. AuthGate will auto-open HomeScreen.
} on FirebaseAuthException catch (e) {
  String msg = 'Registration failed';
  if (e.code == 'email-already-in-use') {
    msg = 'This email is already registered';
  } else if (e.code == 'invalid-email') {
    msg = 'Invalid email address';
  } else if (e.code == 'weak-password') {
    msg = 'Password is too weak';
  } else if (e.code == 'operation-not-allowed') {
    msg = 'Enable Email/Password sign-in in Firebase Console';
  } else {
    msg = e.message ?? e.code;
  }
}

```

```

    }

    setState(() {
      errorMessage = msg;
    });
  } on FirebaseException catch (e) {
    setState(() {
      errorMessage = 'Firestore error: ${e.message ?? e.code}';
    });
  } catch (e) {
    setState(() {
      errorMessage = 'Unexpected error: $e';
    });
  } finally {
    if (mounted) {
      setState(() {
        isLoading = false;
      });
    }
  }
}

```

```

class HomeScreen extends StatefulWidget {
  const HomeScreen({super.key});

  const Divider(),
  const Padding(
    padding: EdgeInsets.symmetric(vertical: 8),
    child: Text(
      'Real-Time Posts',
      style: TextStyle(fontSize: 18, fontWeight: FontWeight.bold),
    ),
  ),
  Expanded(
    child: StreamBuilder<QuerySnapshot<Map<String, dynamic>>>(
      stream: FirebaseFirestore.instance
        .collection('posts')
        .orderBy('createdAt', descending: true)
        .snapshots(),
      builder: (context, snapshot) {
        if (snapshot.connectionState == ConnectionState.waiting) {
          return const Center(child: CircularProgressIndicator());
        }

        if (snapshot.hasError) {
          return Center(
            child: Text('Error: ${snapshot.error}'),
          );
        }
      }
    )
  )
}

```

```

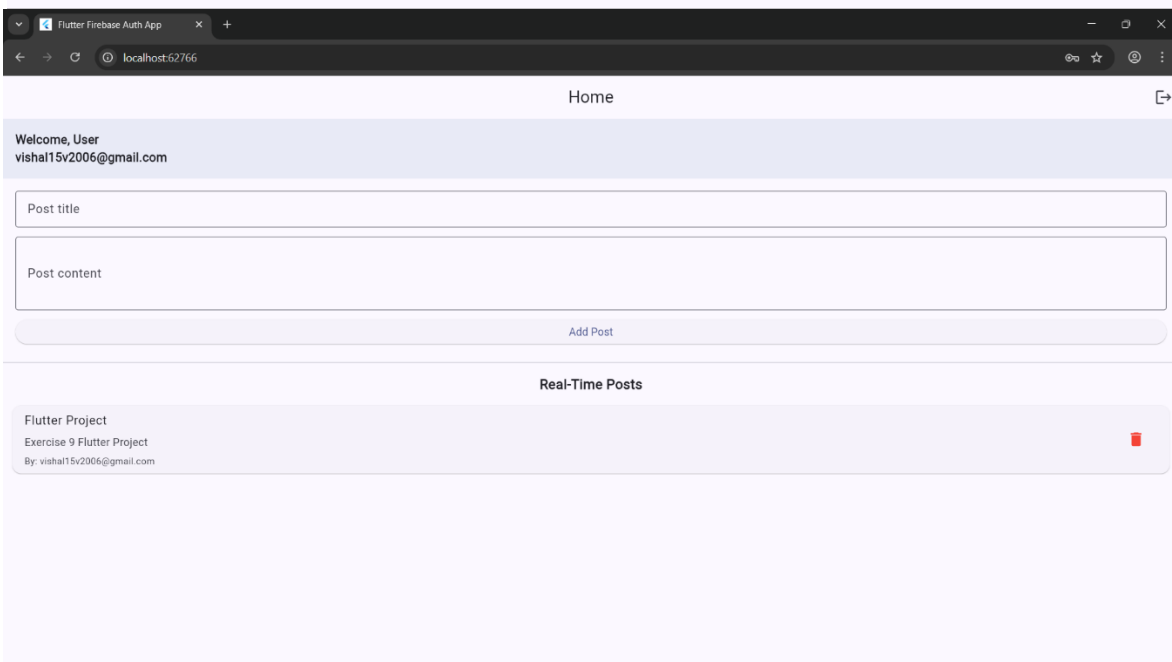
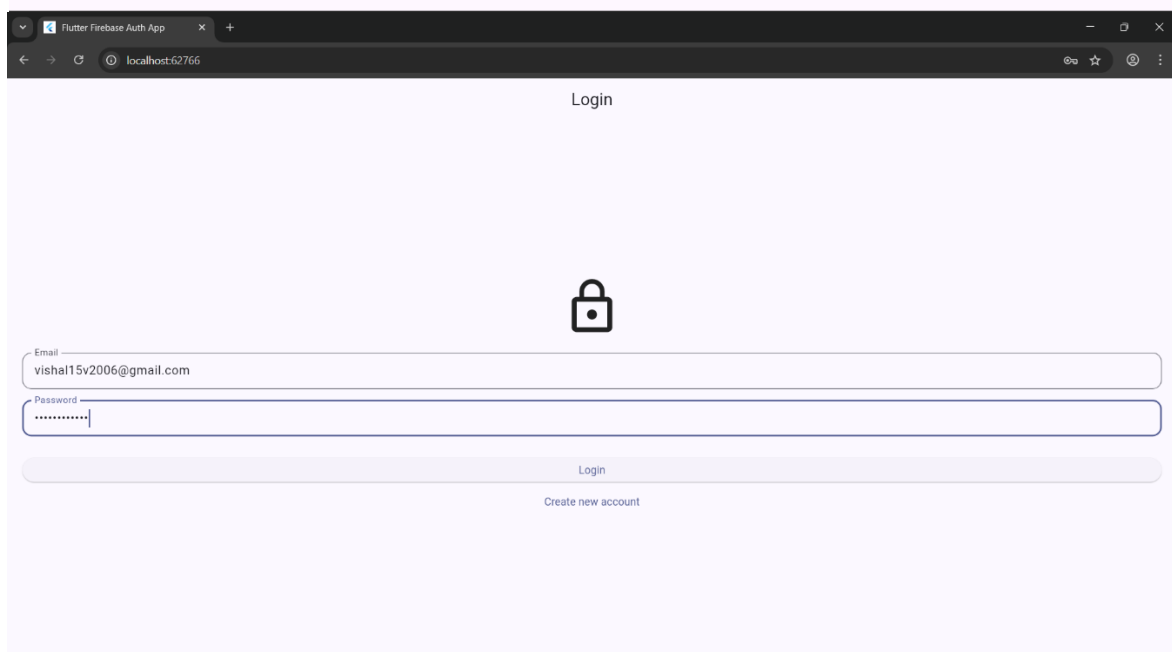
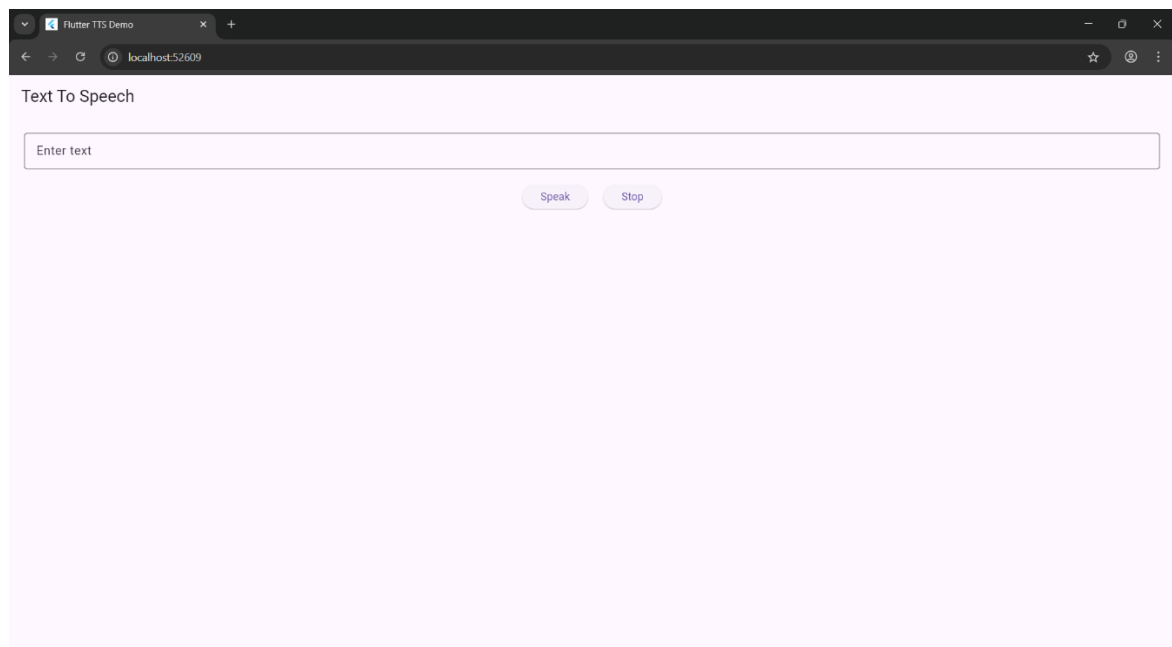
final docs = snapshot.data?.docs ?? [];
if (docs.isEmpty) {
  return const Center(child: Text('No posts yet'));
}

return ListView.builder(
  itemCount: docs.length,
  itemBuilder: (context, index) {
    final doc = docs[index];
    final data = doc.data();

    final isOwner = data['userId'] == user.uid;

    return Card(
      margin: const EdgeInsets.symmetric(
        horizontal: 12,
        vertical: 6,
      ),
      child: ListTile(
        title: Text(data['title'] ?? ''),
        subtitle: Column(
          crossAxisAlignment: CrossAxisAlignment.start,
          children: [
            const SizedBox(height: 6),
            Text(data['content'] ?? ''),
            const SizedBox(height: 6),
            Text(
              'By: ${data['userEmail'] ?? ''}',
              style: const TextStyle(fontSize: 12),
            ),
          ],
        ),
        trailing: isOwner
          ? IconButton(
              onPressed: () => deletePost(doc.id),
              icon: const Icon(Icons.delete, color: Colors.red),
            )
          : null,
      ),
    );
  },
);

```



Flutter Firebase Auth App

localhost:62766

Register



Name

Vishal

Email

vishal15v2006@gmail.com

Password

Register

192.168.0.152

Mail - VISHALV.P - Outlook

Auth - Database - Data - Fire

Flutter Firebase App Setup

VoxStreak - Database - Rules

console.firebase.google.com/u/0/project/auth-a1046/firestore/databases/default/data/~2/users-2fmmroL6tWQHYPEGIEIFr2UVkPvHC3?fb_gclid=Cj0KCQw9-PNBhDFARisABHN6-3Kikw...

Bin - Google Drive

All Bookmarks

 **Firestore**

Search for products

Project Overview

Settings

What's new

Data Connect NEW

Phone Verification NEW

Project shortcuts

Firestore

Authentication

Product categories

Databases and storage

Security

AI services

Hosting and serverless

DevOps and enacoem...

Spark No cost (\$0/month) Upgrade NEW

Auth

Cloud Firestore

Database (default)

(default)

+ Start collection

posts

users

+ Add document

GfQ7yyWqyRSgcT81k8sVPanW2V53

YxS1znVmwKea9U80FKt9zUPRDap2

mmroL6tWQHYPEGIEIFr2UVkPvHC3

mmroL6tWQHYPEGIEIFr2UVkPvHC3

+ Start collection

+ Add field

createdAt: 18 March 2026 at 08:47:28 UTC+5:30

email: "vishal15v2006@gmail.com"

name: "User"

uid: "mmroL6tWQHYPEGIEIFr2UVkPvHC3"

Database location: nam5